

```

14. ALIVE = 'O' # The character representing a living cell.
15. # (!) Try changing DEAD to '.' or another character:
16. DEAD = ' ' # The character representing a dead cell.
17.
18. # (!) Try changing ALIVE to '|' and DEAD to '-'.
19.
20. # The cells and nextCells are dictionaries for the state of the game.
21. # Their keys are (x, y) tuples and their values are one of the ALIVE
22. # or DEAD values.
23. nextCells = {}
24. # Put random dead and alive cells into nextCells:
25. for x in range(WIDTH): # Loop over every possible column.
26.     for y in range(HEIGHT): # Loop over every possible row.
27.         # 50/50 chance for starting cells being alive or dead.
28.         if random.randint(0, 1) == 0:
29.             nextCells[(x, y)] = ALIVE # Add a living cell.
30.         else:
31.             nextCells[(x, y)] = DEAD # Add a dead cell.
32.
33. while True: # Main program loop.
34.     # Each iteration of this loop is a step of the simulation.
35.
36.     print('\n' * 50) # Separate each step with newlines.
37.     cells = copy.deepcopy(nextCells)
38.
39.     # Print cells on the screen:
40.     for y in range(HEIGHT):
41.         for x in range(WIDTH):
42.             print(cells[(x, y)], end='') # Print the # or space.
43.         print() # Print a newline at the end of the row.
44.     print('Press Ctrl-C to quit.')
45.
46.     # Calculate the next step's cells based on current step's cells:
47.     for x in range(WIDTH):
48.         for y in range(HEIGHT):
49.             # Get the neighboring coordinates of (x, y), even if they
50.             # wrap around the edge:
51.             left = (x - 1) % WIDTH
52.             right = (x + 1) % WIDTH
53.             above = (y - 1) % HEIGHT
54.             below = (y + 1) % HEIGHT
55.
56.             # Count the number of living neighbors:
57.             numNeighbors = 0
58.             if cells[(left, above)] == ALIVE:
59.                 numNeighbors += 1 # Top-left neighbor is alive.
60.             if cells[(x, above)] == ALIVE:
61.                 numNeighbors += 1 # Top neighbor is alive.
62.             if cells[(right, above)] == ALIVE:
63.                 numNeighbors += 1 # Top-right neighbor is alive.
64.             if cells[(left, y)] == ALIVE:
65.                 numNeighbors += 1 # Left neighbor is alive.
66.             if cells[(right, y)] == ALIVE:
67.                 numNeighbors += 1 # Right neighbor is alive.
68.             if cells[(left, below)] == ALIVE:

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